

Modern spectroscopic techniques in structural elucidation of organic compounds

Identification of Organic Compounds

- Determining the structures of organic compounds is an important part of organic chemistry.
- Whenever a chemist synthesizes a compound, its structure must be confirmed.
- Scientists search the world for new compounds with physiological activity. If a promising compound is found, its structure needs to be determined.
- Without knowing its structure, chemists cannot design ways to synthesize the compound, nor can they undertake studies to provide insights into its physiological behavior.
- At one time, determining the structure of an organic compound required finding out its **molecular formula by elemental analysis**, determining the compound's physical properties (**its melting point, boiling point, and so on**), and conducting simple chemical tests that indicate the presence (or absence) of certain functional groups.
- Unfortunately, these simple procedures **were inadequate for characterizing molecules with complex structures**, and because a relatively large sample of the compound was needed in order to perform all the tests, they were impractical for the analysis of compounds that were difficult to obtain in large amounts.

Identification of Organic Compounds

Today, a number of different instrumental techniques are used to identify organic compounds. These techniques can be performed quickly on small amounts of a compound and can provide much more information about the compound's structure than simple chemical tests can give.

- **Mass spectrometry** allows us to determine the *molecular mass* and the *molecular formula* of a compound, as well as some of its *structural features*.
- **Infrared (IR) spectroscopy** tells us the *kinds of functional groups* a compound has.
- **Ultraviolet/Visible (UV/Vis) spectroscopy** provides information about organic compounds with conjugated double bonds.
- **Nuclear magnetic resonance (NMR) spectroscopy** provides information about the carbon–hydrogen framework of an organic compound.

Ultraviolet And Visible Spectroscopy

Ultraviolet and visible (UV/Vis) spectroscopy provides information about compounds that have conjugated double bonds. Ultraviolet light and visible light have just the right energy to cause an electronic transition in a molecule—that is, to promote an electron from one molecular orbital to another of higher energy.

Depending on the energy needed for the electronic transition, a molecule will absorb either ultraviolet or visible light. If it absorbs **ultraviolet light**, a UV spectrum is obtained; if it absorbs **visible light**, a visible spectrum is obtained. *Ultraviolet light* has wavelengths ranging from 180 to 400 nm (nanometers); *visible light* has wavelengths ranging from 400 to 780 nm.

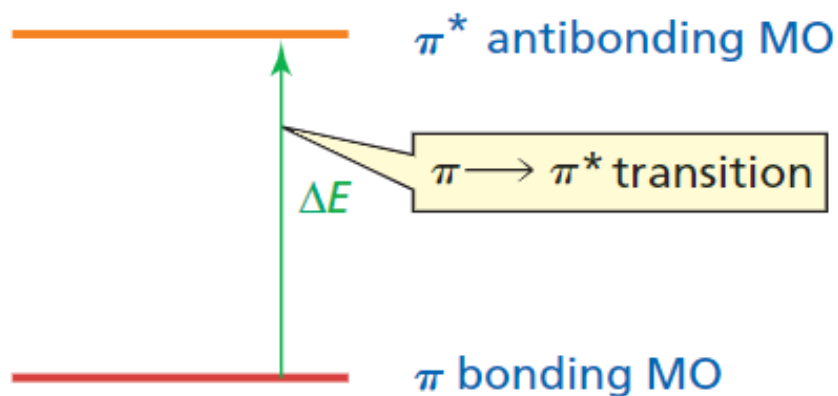
Wavelength (λ) is inversely related to the energy of the radiation, so the shorter the wavelength, the greater the energy of the radiation. Ultraviolet light, therefore, has greater energy than visible light.

The shorter the wavelength, the greater the energy of the radiation.

$$E = \frac{hc}{\lambda}$$

Ultraviolet And Visible Spectroscopy

- In the ground-state electronic configuration of a molecule, all the electrons are in the lowest-energy molecular orbitals.
- When a molecule absorbs light with sufficient energy to **promote an electron** to a **higher-energy** molecular orbital—that is, when it undergoes an **electronic transition**—the molecule is then in **an excited state**.
- Ultraviolet and visible light has just enough energy to promote an electron from a π *bonding* MO to a π^* *antibonding* MO. Therefore, **only compounds with π bonds can produce UV/Vis spectra**.



UV Spectrum

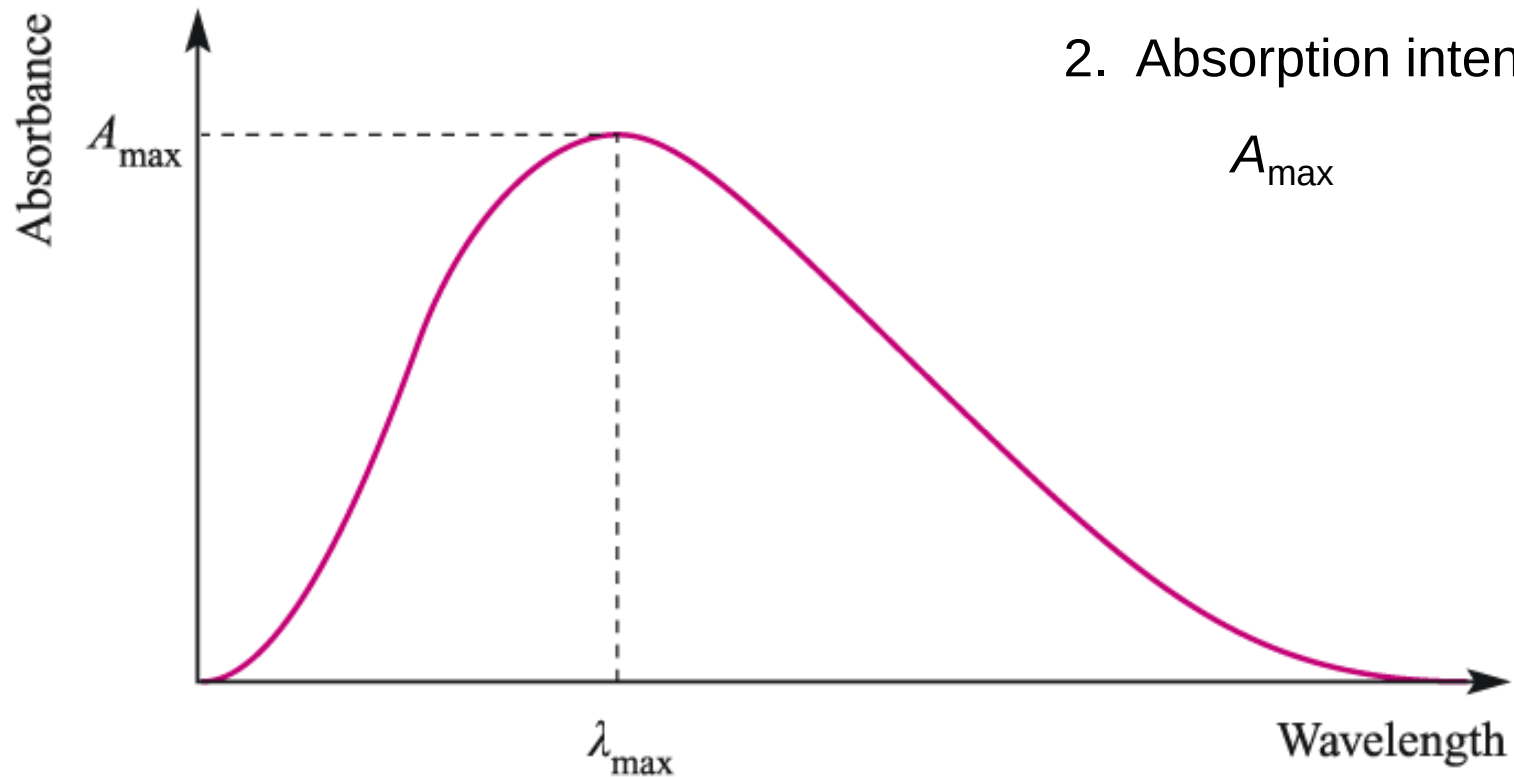
The two main properties of an absorbance peak are:

1. Absorption wavelength

λ_{max}

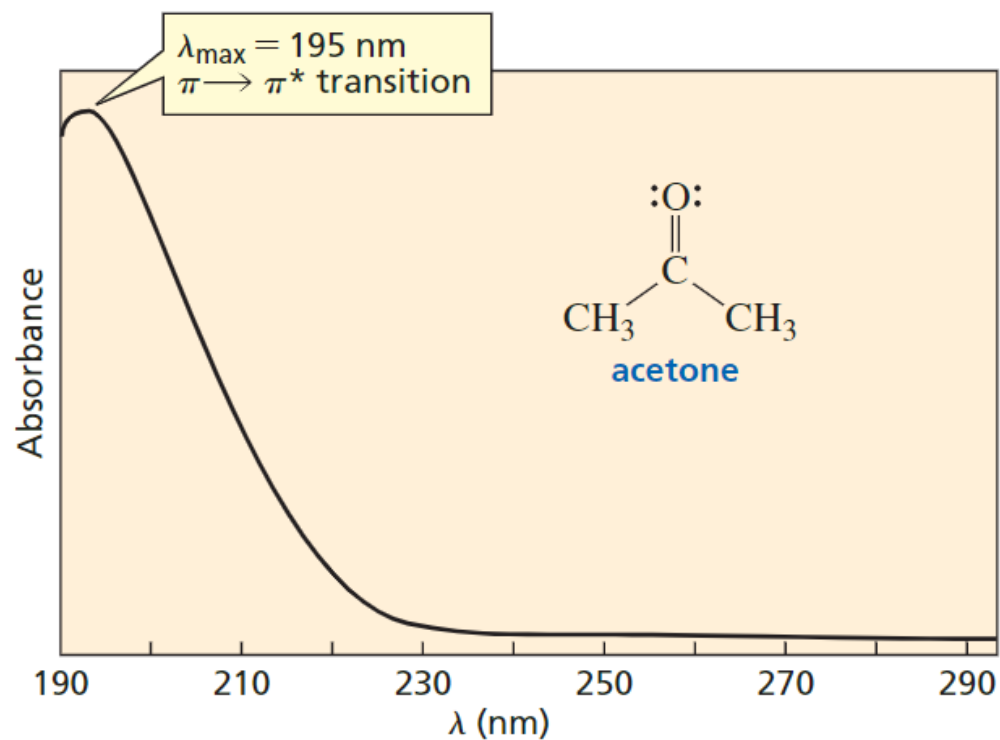
2. Absorption intensity

A_{max}



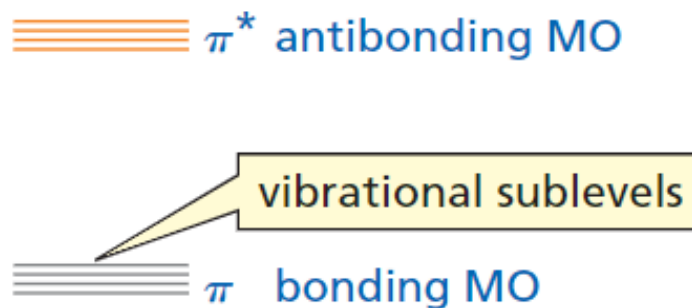
Ultraviolet And Visible Spectroscopy

The UV spectrum of acetone is shown in Figure . The λ_{max} (stated as “lambda max”) is the wavelength at which the absorption band for the $\pi \rightarrow \pi^*$ transition (stated as “ π to π^* ”) has its maximum absorbance. For acetone, $\lambda_{\text{max}} = 195 \text{ nm}$.

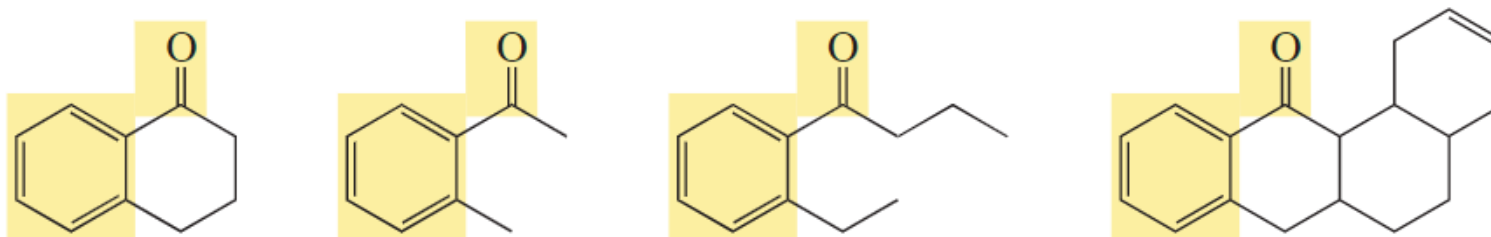


Ultraviolet And Visible Spectroscopy

The absorption band is **broad** because each electronic state has vibrational sublevels and electronic transitions can occur from and to these different sublevels. As a result, the electronic transitions span a range of wavelengths.



A **chromophore** is the part of a molecule that absorbs UV or visible light. The carbonyl group is the chromophore of acetone. The following four compounds all have the same chromophore, so they all have approximately the same λ_{max} .



Ultraviolet And Visible Spectroscopy

Beer–Lambert law

Wilhelm Beer and Johann Lambert independently proposed that the absorbance of a sample at a given wavelength depends on the amount of absorbing species that the light encounters as it passes through a solution of the sample. In other words, absorbance depends on both the concentration of the sample and the length of the light path through the sample. The relationship between absorbance, concentration, length of the light path, and molar absorptivity, known as the **Beer–Lambert law**, is given by

$$A = c l \varepsilon, \text{ where}$$

A = absorbance of the sample

c = concentration of the sample, in moles/liter

l = length of the light path through the sample, in centimeters

ε = molar absorptivity ($\text{M}^{-1}\text{cm}^{-1}$)

The **molar absorptivity** (ε) is a constant that is characteristic of the compound at a particular wavelength. It is the absorbance that would be observed for a 1.00 M solution in a cell with a 1.00-cm path length. (The abbreviation ε comes from the fact that molar absorptivity was formerly called the extinction coefficient.)

Beer-Lambert Law

Beer-Lambert Law:

$$\log(I_0/I) = \epsilon lc$$

$$\epsilon = A/cl$$

$$A = \epsilon lc$$

$$A = \epsilon C \text{ (when } l \text{ is 1 cm)}$$

I_0 = intensity of incident light

I = intensity of transmitted light

ϵ = molar absorptivity coefficient in $\text{cm}^2 \text{ mol}^{-1}$

c = concentration in mol L^{-1}

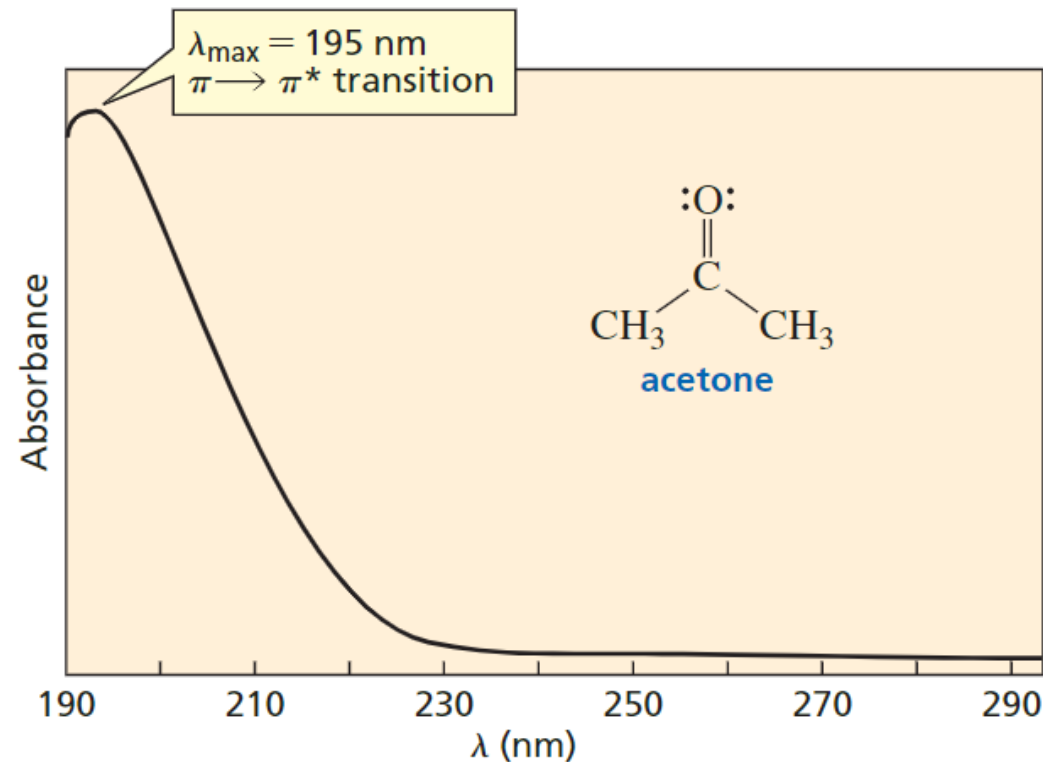
l = path length of absorbing solution in cm

A = absorbance = $\log(I_0/I)$

Ultraviolet And Visible Spectroscopy

Beer–Lambert law

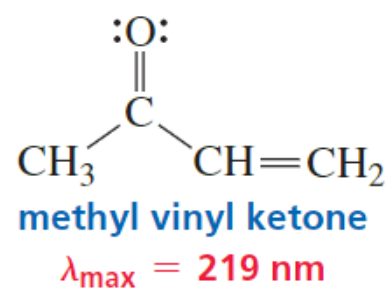
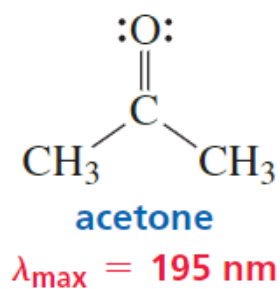
For example, the molar absorptivity of acetone dissolved in hexane is $9000 \text{ M}^{-1} \text{ cm}^{-1}$ at 195 nm. The solvent in which the sample is dissolved is reported because molar absorptivity is not exactly the same in all solvents. Therefore, the UV spectrum of acetone in hexane would be reported as $\lambda_{\text{max}} = 195 \text{ nm}$ ($\epsilon_{\text{max}} = 9000$, hexane).



Ultraviolet And Visible Spectroscopy

THE EFFECT OF CONJUGATION ON λ_{max}

The wavelength at which the $\pi \rightarrow \pi^*$ transition occurs increases as the number of conjugated double bonds in the compound increases. For example, the λ_{max} for methyl vinyl ketone is at a longer wavelength (219 nm) than the λ_{max} for acetone (195 nm), because methyl vinyl ketone has two conjugated double bonds, whereas acetone has only one double bond.

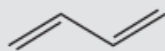
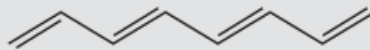
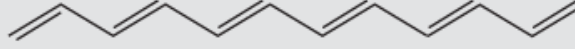


Ultraviolet And Visible Spectroscopy

THE EFFECT OF CONJUGATION ON $\lambda_{\max}$

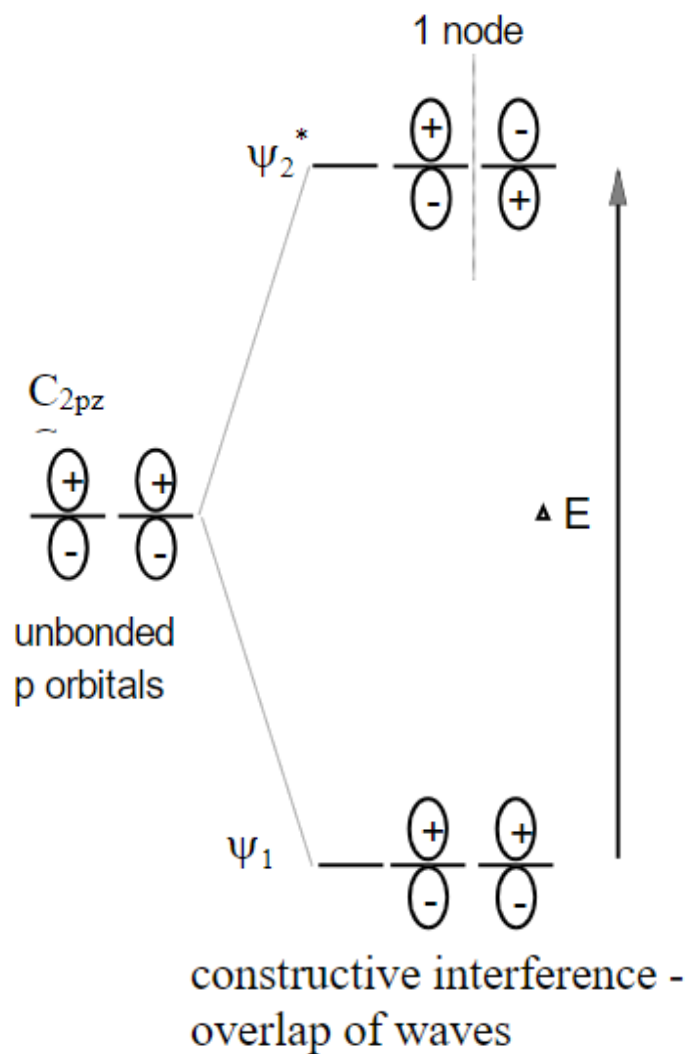
The $\lambda_{\max}$ values of the $\pi \rightarrow \pi^*$ transition for several conjugated polyenes are listed in Table 14.6. Thus, the $\lambda_{\max}$ of a compound can be used to estimate the number of conjugated double bonds in a compound. Notice that both the $\lambda_{\max}$ and the molar absorptivity increase as the number of conjugated double bonds increases.

Table 14.6 Values of $\lambda_{\max}$ and ϵ for Ethylene and Conjugated Polyenes

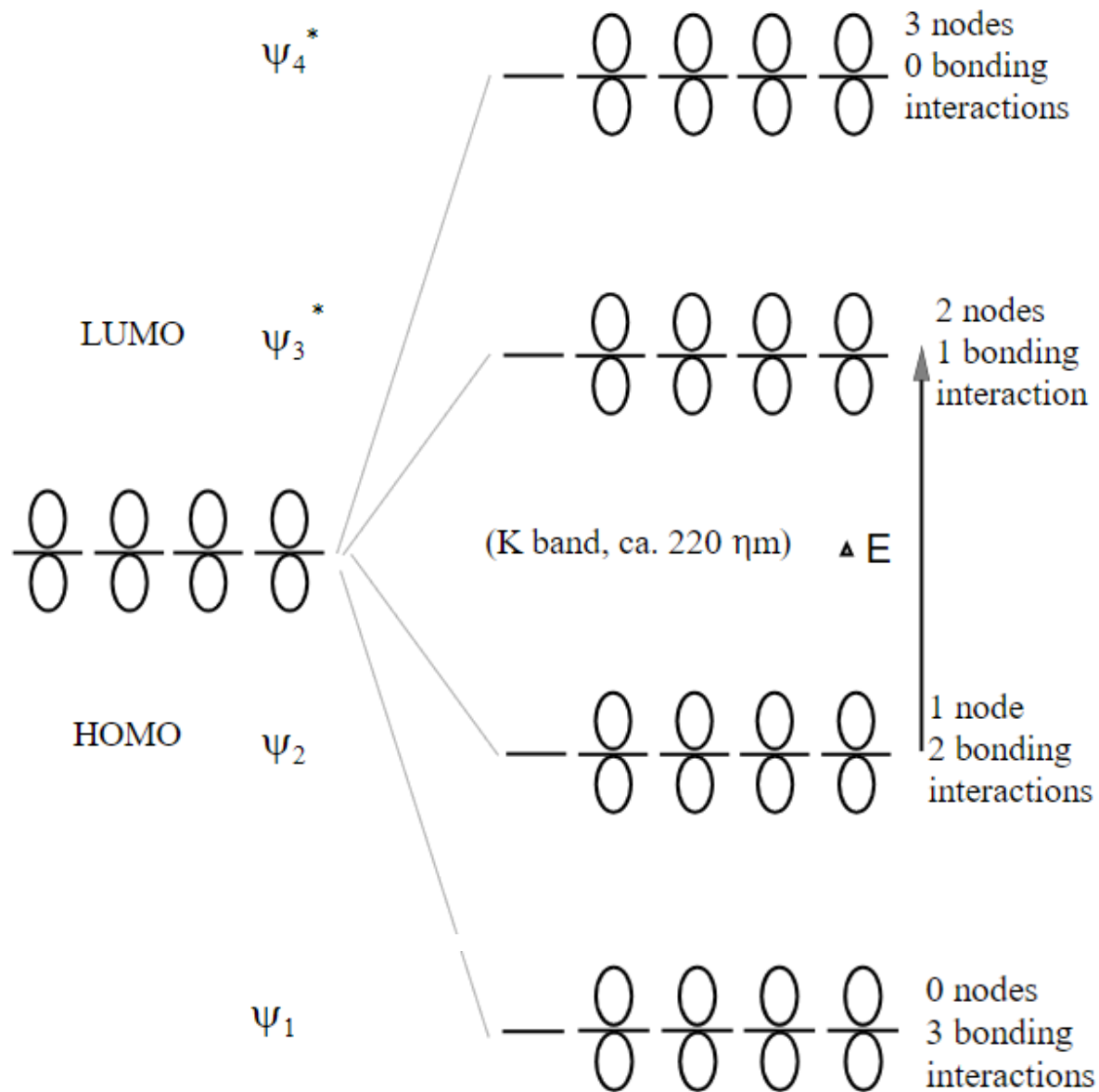
Compound	$\lambda_{\max}$ (nm)	ϵ ($\text{M}^{-1} \text{cm}^{-1}$)
$\text{H}_2\text{C}=\text{CH}_2$	165	15,000
	217	21,000
	256	50,000
	290	85,000
	334	125,000
	364	138,000

The $\lambda_{\max}$ increases as the number of conjugated double bonds increases.

EFFECT OF CONJUGATION



ethylene

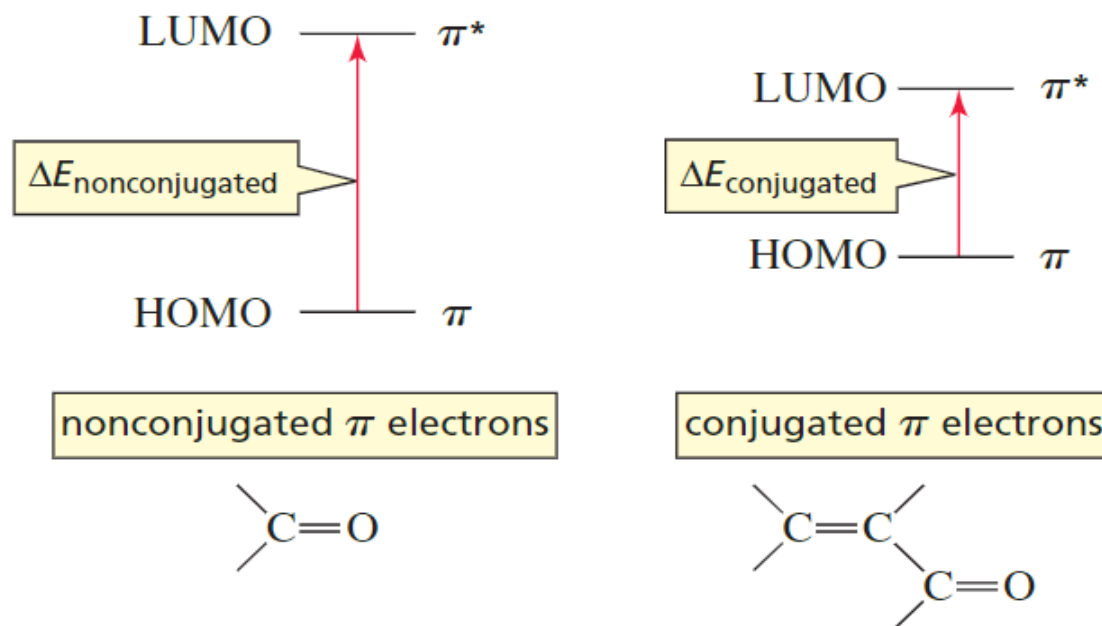


1,3-butadiene

Ultraviolet And Visible Spectroscopy

THE EFFECT OF CONJUGATION ON λ_{max}

Conjugation raises the energy of the **HOMO (highest occupied molecular orbital)** and lowers the energy of the **LUMO (lowest unoccupied molecular orbital)**, so less energy is required for an electronic transition in a conjugated system than in a nonconjugated system. Therefore, the more conjugated double bonds in a compound, the less energy required for the electronic transition so the longer the wavelength at which it occurs.



Calculation of λ_{max} of Organic Compounds Using Woodward Fieser Rules

In 1945 **Robert Burns Woodward** gave certain rules for correlating λ_{max} with molecular structure. In 1959 **Louis Frederick Fieser** modified these rules with more experimental data, and the modified rule is known as **Woodward-Fieser Rules**. It is used to calculate the position and λ_{max} for a given structure by relating the position and degree of substitution of chromophore.



Robert Burns Woodward



Louis Frederick Fieser

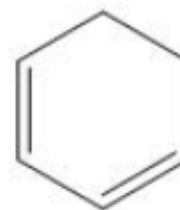
WOODWARD- FIESER RULES:

Each type of diene or triene system is having a certain fixed value at which absorption takes place; this constitutes the **Base value or Parent value**. The contribution made by various alkyl substituents or ring residue, double bond extending conjugation and polar groups such as $-\text{Cl}$, $-\text{Br}$ etc are added to the basic value to obtain λ_{max} for a particular compound.

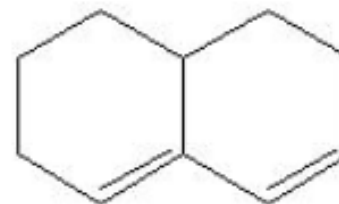
Calculation of λ_{max} of Organic Compounds Using Woodward Fieser Rules

I) CONJUGATED DIENE CORRELATIONS:

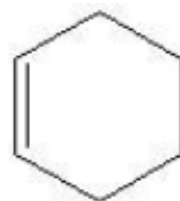
a) Homoannular Diene:- Cyclic diene having conjugated double bonds in same ring.



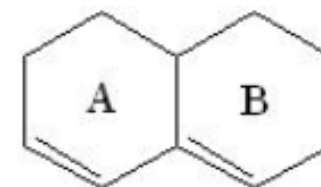
b) Heteroannular Diene:- Cyclic diene having conjugated double bonds in different rings.



c) Endocyclic double bond:- Double bond present in a ring.



d) Exocyclic double bond:- Double bond in which one of the doubly bonded atoms is a part of a ring system.



Here Ring A has one exocyclic and endocyclic double bond. Ring B has only one endocyclic double bond.

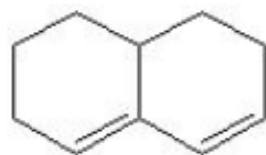
Calculation of $\lambda_{\max}$ of Organic Compounds Using Woodward Fieser Rules

PARENT VALUES AND INCREMENTS FOR DIFFERENT SUBSTITUENTS/GROUPS:

I) CONJUGATED DIENE CORRELATIONS:

- i) Base value for homoannular diene = 253 nm
- ii) Base value for heteroannular diene = 214 nm
- iii) Alkyl substituent or Ring residue attached to the parent diene = 5 nm
- iv) Double bond extending conjugation = 30 nm
- v) Exocyclic double bonds = 5 nm
- vi) Polar groups:
 - a) -OAc = 0 nm
 - b) -OAlkyl = 6 nm
 - c) -Cl, -Br = 5 nm

Eg:



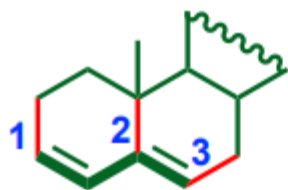
Base value = 214 nm

Ring residue = $3 \times 5 = 15$ nm

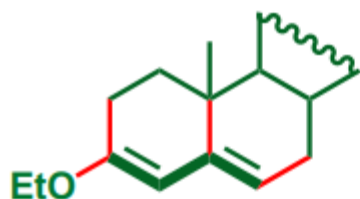
Exocyclic double bond = $1 \times 5 = 5$ nm

$$\lambda_{\max} = 214 + 15 + 5 = 234 \text{ nm}$$

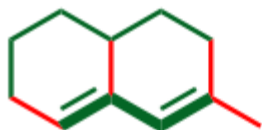
Calculation of λ_{max} of Organic Compounds Using Woodward Fieser Rules



Transoid (base):	214 nm
3 ring residues :	+15
1 exocyclic C=C:	+ 5
Total:	234 nm
Observed:	235 nm

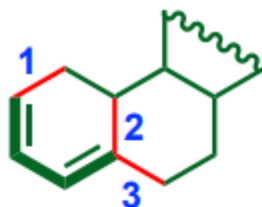


Transoid (base):	214 nm
3 ring residues:	+15
1 exocyclic C=C:	+ 5
-OR:	+ 6
Total:	240 nm
Observed:	241 nm

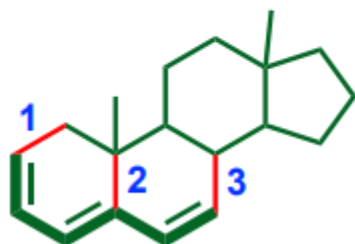


Transoid (base):	214 nm
3 Ring residues:	+15
1 Alkyl substituent:	+ 5
1 Exocyclic C=C:	+ 5
Total:	239 nm

Calculation of λ_{max} of Organic Compounds Using Woodward Fieser Rules



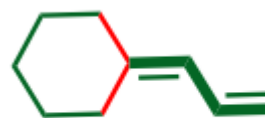
Cisoid (base):	253 nm
3 ring residues:	+15
1 exocyclic C=C:	+ 5
Total:	273 nm
Observed:	275 nm



Cisoid (base):	253 nm
3 Ring residues:	+15
1 Exocyclic C=C:	+ 5
Double-bond Extending Conjugation:	+30
Total:	303 nm
Observed:	304 nm



Base value:	214 nm
2 Ring residue	+10
Exocyclic C=C:	+ 5
Total:	229 nm
Observed:	230 nm



Base value:	214 nm
2 Ring residue:	+10
Exocyclic C=C:	+ 5
Total	229 nm
Observed:	236 nm

Calculation of λ_{max} of Organic Compounds Using Woodward Fieser Rules

Rules of Enone & Dienone Absorption

Base values:

➤ Acyclic α,β -unsaturated ketones	215 nm
➤ 6-membered cyclic α,β -unsaturated ketones	215 nm
➤ 5-membered cyclic α,β -unsaturated ketones	202 nm
➤ α,β -unsaturated aldehydes	210 nm
➤ α,β -unsaturated carboxylic acid & esters	195 nm

Increments for:

Double bond extending conjugation (DEC):	+30
Exocyclic double bond:	+ 5
Homodiene component:	+39

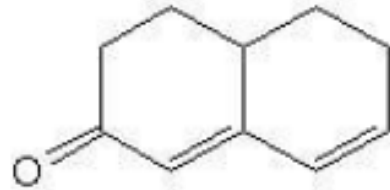
Calculation of λ_{max} of Organic Compounds Using Woodward Fieser Rules

Increments for:

Alkyl group/ring residue: α		+10
β		+12
γ & higher		+18
Polar groups: -OH:	α	+35
	β	+30
	δ	+50
	-OAc: α, β, γ	+ 6
-OMe:	α	+35
	β	+30
	γ	+17
	δ	+31
-SAlk:	β	+85
-Cl:	α	+15
	β	+12
-Br:	α	+25
	β	+30
-NR ₂ :	β	+95

Calculation of λ_{max} of Organic Compounds Using Woodward Fieser Rules

Eg:



Base value = 214 nm

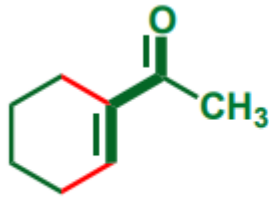
β - Substituents = $1 \times 12 = 12$ nm

δ - Substituents = $1 \times 18 = 18$ nm

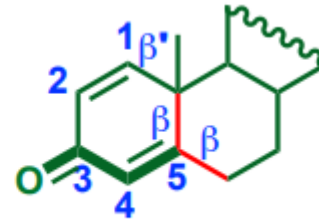
Double bond extending conjugation = $1 \times 30 = 30$ nm

Exocyclic double bond = 5 nm

$\lambda_{\text{max}} = 279$ nm

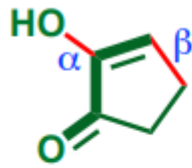


Base value:	215 nm
α substituent:	+10
β substituent:	+12
Total:	<u>237 nm</u>
Observed:	232 nm

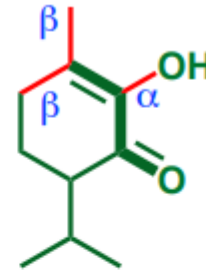


$\Delta^{4,5}$ system (base):	215 nm
2 β substituents:	+24
1 exocyclic C=C:	+ 5
Total:	<u>244 nm</u>
Observed:	245 nm

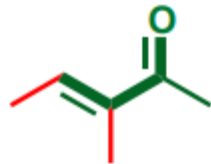
Calculation of λ_{max} of Organic Compounds Using Woodward Fieser Rules



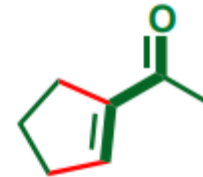
Base value:	202 nm
β substituent:	+12
α -OH:	+35
Total:	249 nm
Observed:	247 nm



Base value:	215 nm
2 β substituents:	+24
α -OH:	+35
Total:	274 nm
Observed:	270 nm

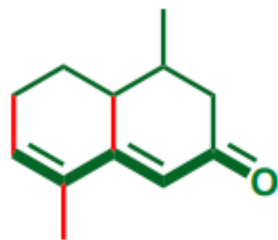


Base value:	215 nm
α alkyl substituent:	+10
β alkyl substituent:	+12
Total:	237 nm

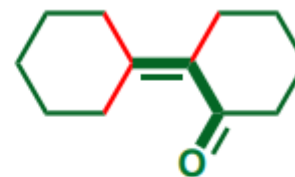


Base value:	215 nm
α alkyl:	+10
β alkyl:	+12
Total:	237 nm

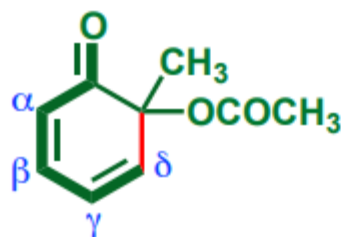
Calculation of λ_{max} of Organic Compounds Using Woodward Fieser Rules



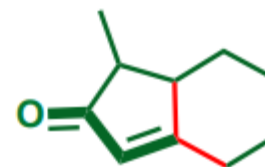
Base value:	215 nm
1 DEC:	+30
Exocyclic C=C:	+ 5
β -alkyl substituent:	+12
γ -alkyl substituent:	+18
δ -alkyl substituent:	+18
Total:	<u>298 nm</u>



Base value:	215 nm
1 α -alkyl:	+10
2 β -alkyl:	+24
2 Exocyclic C=C:	+10
Total:	<u>259 nm</u>



Base value:	215 nm
1 DEC:	+30
Homocyclic diene:	+39
δ ring residue:	+18
Total:	<u>302 nm</u>
Observed:	300 nm

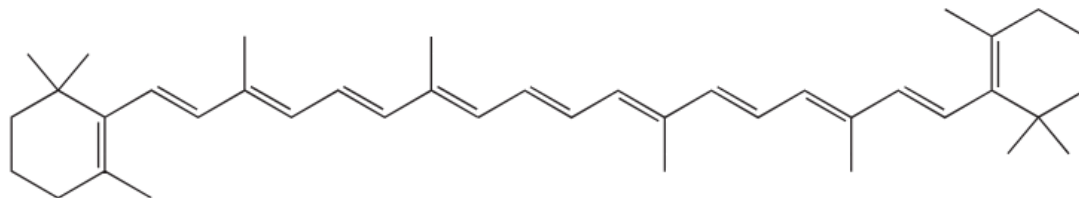


Base value:	202 nm
Exocyclic C=C:	+ 5
2 β -ring residues:	+24
Total:	<u>231 nm</u>
Observed:	226 nm

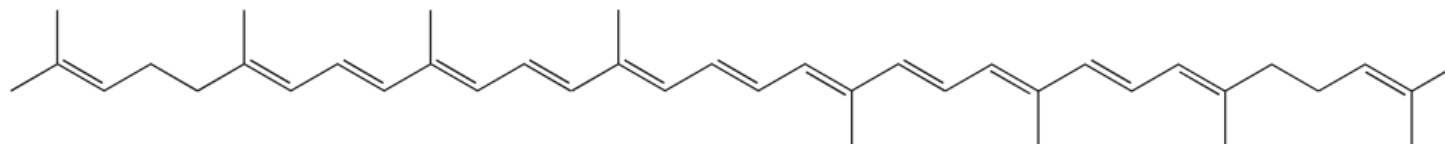
Ultraviolet And Visible Spectroscopy

THE EFFECT OF CONJUGATION ON λ_{max}

If a compound has enough conjugated double bonds, it will absorb visible light (light with wavelengths $> 400\text{ nm}$) and the compound will be colored. For example, β -carotene, a precursor of vitamin A with a $\lambda_{\text{max}} = 455\text{ nm}$, is an orange substance found in carrots, apricots, and the feathers of flamingos. Lycopene with a $\lambda_{\text{max}} = 474\text{ nm}$ —found in tomatoes, watermelon, and pink grapefruit—is red.



β -carotene
 $\lambda_{\text{max}} = 455\text{ nm}$

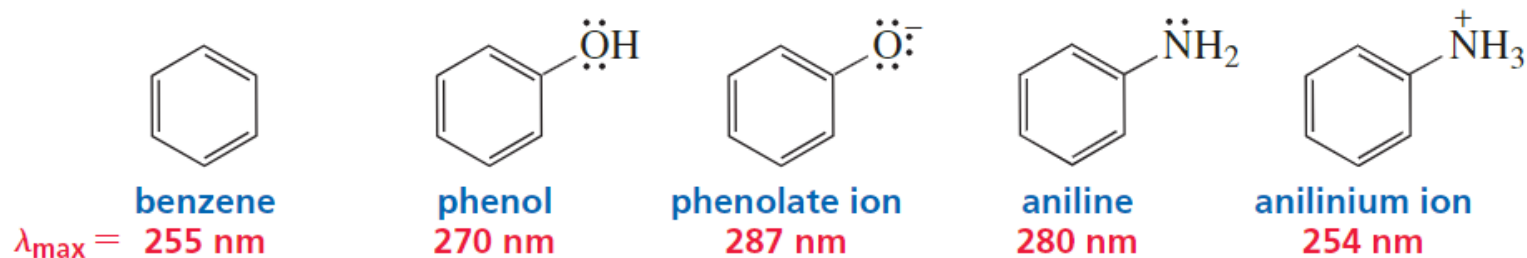


lycopene
 $\lambda_{\text{max}} = 474\text{ nm}$

Ultraviolet And Visible Spectroscopy

THE EFFECT OF CONJUGATION ON $\lambda_{\max}$

An **auxochrome** is a substituent that, when attached to a chromophore, alters both the $\lambda_{\max}$ and the intensity of the absorption, usually increasing both. For example, OH and NH_2 groups are auxochromes. The lone-pair electrons on oxygen and nitrogen in the compounds shown here are available to interact with the π electron cloud of the benzene ring; such an interaction increases $\lambda_{\max}$.



Removing a proton from phenol increases the $\lambda_{\max}$ because the phenolate ion has an additional lone pair. Protonating aniline decreases the $\lambda_{\max}$ because the lone pair is no longer available to interact with the π cloud of the benzene ring. Because the anilinium ion does not have an auxochrome, its $\lambda_{\max}$ is similar to that of benzene.

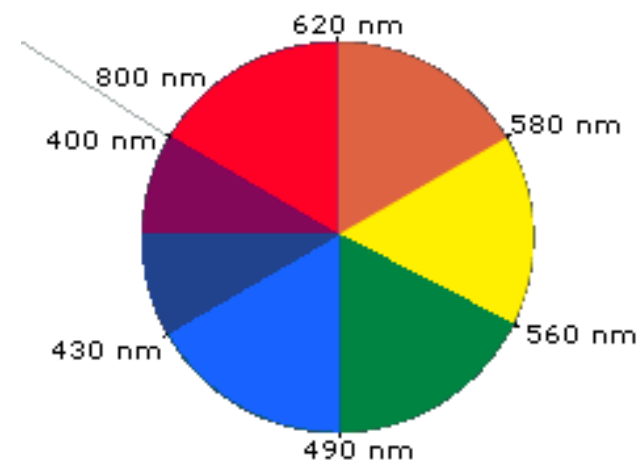
THE VISIBLE SPECTRUM AND COLOR

- White light is a mixture of all visible wavelengths. If any **of these wavelengths are removed from white light**, the eye registers **the remaining light as colored**.
- Therefore, any compound that absorbs visible light appears colored. The perceived color depends on the precise wavelengths reaching the eye. The **wavelengths that the compound does not absorb are reflected back to the viewer**, producing the color the viewer sees.

The relationship between the wavelengths of the light that a **substance absorbs and the substance's observed color** is shown in Table below.

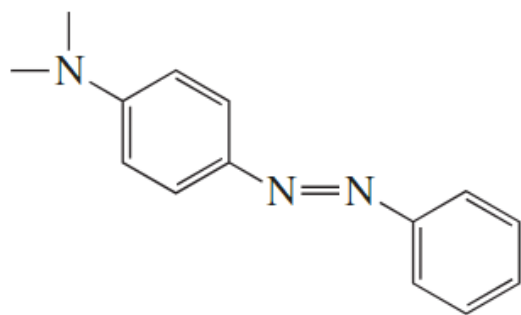
Dependence of the Color Observed on the Wavelength of Light Absorbed

Wavelengths absorbed (nm)	Color absorbed	Color observed
380–460	blue-violet	yellow
380–500	blue	orange
440–560	blue-green	red
480–610	green	purple
540–650	orange	blue
380–420 and 610–700	purple	green

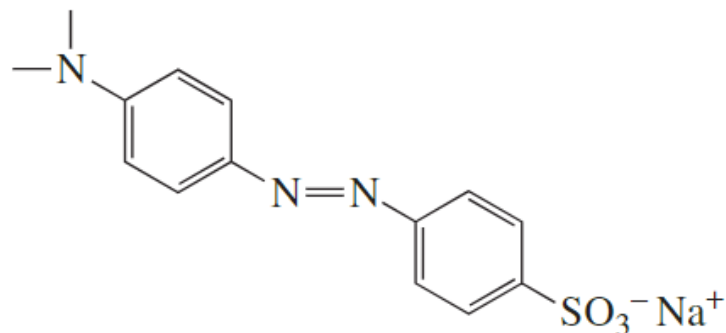


THE VISIBLE SPECTRUM AND COLOR

Azobenzenes (benzene rings connected by an $\text{N}=\text{N}$ bond) have an extended conjugated system that causes them to absorb visible light. The two shown here are used commercially as dyes. Changing the number of conjugated double bonds and the substituents attached to them creates a large number of different colors. Notice that the only difference between butter yellow and methyl orange is an SO_3^-Na^+ group.



butter yellow
an azobenzene



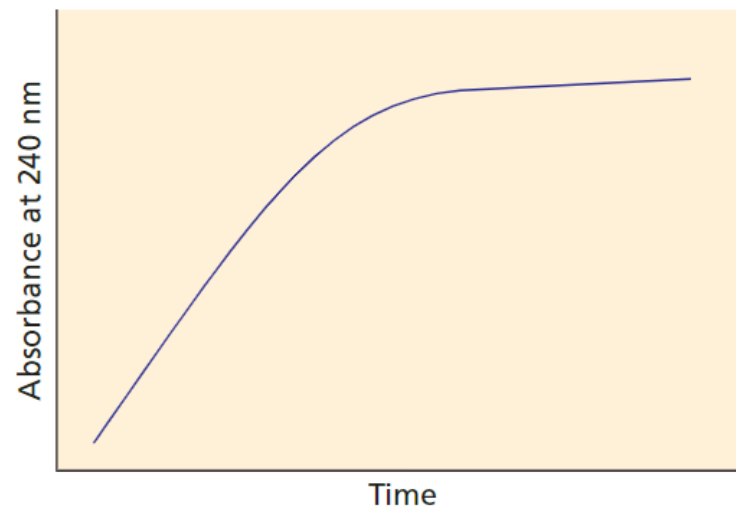
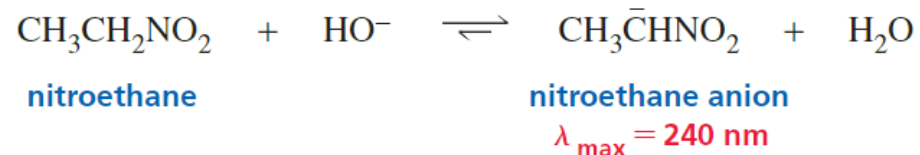
methyl orange
an azobenzene

SOME USES OF UV/VIS SPECTROSCOPY

UV/Vis spectroscopy is not nearly as useful as other instrumental techniques for determining the structures of organic compounds. However, UV/Vis spectroscopy has many other important uses.

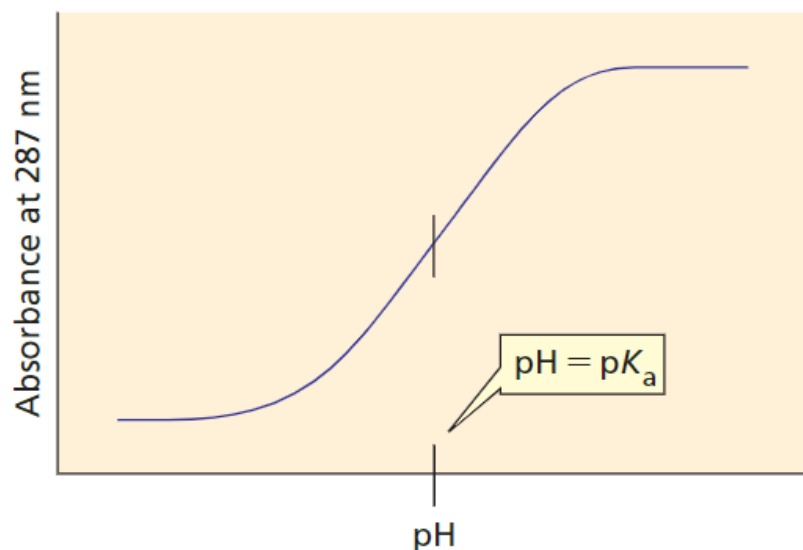
UV/Vis spectroscopy is often used to measure reaction rates. The rate of any reaction can be measured, as long as one of the reactants or one of the products absorbs UV or visible light at a wavelength at which the other reactants and products have little or no absorbance.

For example, the anion of nitroethane has a λ_{max} at 240 nm, but neither the other product (H_2O) nor the reactants show any significant absorbance at this wavelength. In order to measure the rate at which hydroxide ion removes a proton from nitroethane to form the nitroethane anion, the UV spectrophotometer is adjusted to measure absorbance at 240 nm as a function of time instead of absorbance as a function of wavelength



SOME USES OF UV/VIS SPECTROSCOPY

The pK_a of a compound can be determined by UV/Vis spectroscopy if either the acidic form or the basic form of the compound absorbs UV or visible light. For example, the phenolate ion has a $\lambda_{\max}$ at 287 nm. If the absorbance at 287 nm is monitored as a function of pH, the pK_a of phenol can be ascertained by determining the pH at which exactly one-half the increase in absorbance has occurred. At this pH, half of the phenol has been converted into phenolate ion, so this pH is equal to the pK_a of the compound.



Because the equilibrium between the acid and its anion may be described by

$$\log K_a = \log[\text{H}_3\text{O}^+] + \log \frac{[\text{A}^-]}{[\text{HA}]}$$